

Malak Mostafa

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SUMMARY

Computer Science and Artificial Intelligence student at Capital University with a foundation in programming, electronics, and technology, with hands-on experience building electronics projects and web applications, proficient in multiple programming languages and web technologies, passionate about building impactful software solutions, fast learner eager to develop new skills, gain practical experience, and apply academic knowledge.

EDUCATION

Bachelor of Science - Computer Science and Artificial Intelligence

Capital University, Egypt

PROJECTS

Course manager website

Designed and developed a frontend interface for a course organizer, built with HTML, CSS

Animal encyclopedia website

Participated in building an interactive animal encyclopedia website featuring categorized animal profiles with images and detailed information, and personalized user profiles with client-side data persistence using localStorage to save favorites and posts; responsible for the Sign Up and Login pages including form validation, and led the UI design for the majority of the site's pages. Built with HTML, CSS, and JavaScript

Electronics Lab Projects — Breadboard & Proteus Simulation | Circuit Design • Electronics

- Built basic resistive circuits with fixed and variable resistors, applying Ohm's Law to control current flow and LED/bulb brightness
- Designed a **Slow Light Bulb** circuit using a capacitor to demonstrate RC charging and discharging behavior, producing gradual turn-on and turn-off of a bulb
- Implemented a **Blinking LEDs** circuit using a 555 Timer IC in astable mode with resistors and capacitors to generate a continuous periodic signal driving simultaneously blinking LEDs
- Constructed a **Buzzer Activation** circuit using two transistors, two diodes, and a capacitor to trigger a buzzer when a switch is closed, demonstrating transistor switching behavior
- Built a **Rain Detector** circuit using a 555 Timer IC, a water-sensitive sensor, and a transistor to activate a buzzer upon water contact with the sensor

SKILLS

Languages: C • Python • JavaScript

Web Technologies: HTML5 • CSS3

Electronics: Circuit Design • Breadboard Prototyping • Proteus Simulation

KEY STRENGTHS

Fast Learner • Self-Motivated • Curious & Research-Oriented • Attention to Detail • Team Collaboration

LANGUAGES

Arabic — Native | English — C1 (Advanced)